

**No.    My SQL Logical Questions**

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Salary:- Maximum salary

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Salary:- Nth Highest salary

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Salary:- Top Nth salary

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salary + department:- Department-wise Total Salary

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salary + department:- Highest Salary ka Department kaun sa hai?

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salary + department:-Department Having Total Salary > 100000

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salary + department:- Department-wise average salary?

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value count:- Email

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Duplicate values in a Table?

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Duplicate values remove

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Replace a column value:- M to F & F to M

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## Find Maximum Salary

| emp_name | salary |
|----------|--------|
| AMIT     | 50000  |
| ROHIT    | 90000  |
| NEHA     | 90000  |
| PRIYA    | 65000  |

- Sirf salary chahiye

```
SELECT MAX(salary) AS highest_salary FROM employees;
```

| max_salary |
|------------|
| 90000      |

- Employee with Maximum Salary (Name + Salary chahiye)

```
SELECT emp_name, salary FROM employees
WHERE salary = (
    SELECT MAX(salary) FROM employees
);
```

| emp_name | salary |
|----------|--------|
| ROHIT    | 90000  |
| NEHA     | 90000  |

## Find 3rd Highest Salary

| emp_name | salary |
|----------|--------|
| AMIT     | 50000  |
| ROHIT    | 90000  |
| NEHA     | 70000  |
| PRIYA    | 80000  |
| RAVI     | 60000  |

### Using Limit

- The limit clause has two components, the **First component** is to skip a number of rows from the top and the **second component** is to display the number of rows we want.

```
-- Syntex:- Using Limit
Select DISTINCT Salary from table_name order by Salary DESC limit n-1,1;

-- 3rd highest salary
SELECT salary FROM employees ORDER BY salary DESC LIMIT 2,1;
SELECT emp_name, salary FROM employees ORDER BY salary DESC LIMIT 2,1;

-- (OR) Standard style
SELECT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 2;

-- For DISTINCT duplicate salary ko handle karega
SELECT DISTINCT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 2;

-- Example:- 4th Highest salary using limit
Select DISTINCT emp_name, salary from Employee order by salary DESC limit 3,1;
```

### Using Subquery

```
SELECT MAX(salary) AS third_highest_salary
FROM employees
WHERE salary < (
    SELECT MAX(salary)
    FROM employees
    WHERE salary < (
        SELECT MAX(salary)
        FROM employees
    )
);
```

```
+-----+  
| third_highest_salary |  
+-----+  
| 70000 |  
+-----+
```

## Employee name ke sath 3rd Highest Salary

- Using Sub Query

```
SELECT emp_name, salary
FROM employees
WHERE salary = (
    SELECT MAX(salary)
    FROM employees
    WHERE salary < (
        SELECT MAX(salary)
        FROM employees
        WHERE salary < (
            SELECT MAX(salary)
            FROM employees
        )
    )
);
```

| emp_name | salary |
|----------|--------|
| NEHA     | 70000  |

- Agar Duplicate hai to dono print honge

| emp_name | salary |
|----------|--------|
| NEHA     | 70000  |
| RAVI     | 70000  |

## Using DENSE\_RANK() (Most Preferred for Experienced Interview)

### Using COUNT() Subquery

## Find Top n Salaries

### Find Top 4 Salaries

- Using Limit

```
SELECT DISTINCT salary FROM employees ORDER BY salary DESC LIMIT 4;

-- with name isme distinct nahi aayega
SELECT emp_name, salary FROM employees ORDER BY salary DESC LIMIT 4;
```

- DISTINCT salaries ke saath employees ke names bhi chahiye to using Subquery

```
SELECT emp_name, salary
FROM employees
WHERE salary IN (
    SELECT DISTINCT salary
    FROM employees
    ORDER BY salary DESC
    LIMIT 4
)
ORDER BY salary DESC;
```

- Using Subquery

```
SELECT emp_name, salary
FROM (
    SELECT emp_name,
           salary,
           DENSE_RANK() OVER(ORDER BY salary DESC) AS rnk
    FROM employees
) AS temp
WHERE rnk <= 4;

-- In Oracle
SELECT SAL FROM(SELECT DISTINCT SAL FROM EMP WHERE SAL IS NOT NULL ORDER BY SAL
DESC)WHERE ROWNUM <6;
```

## Department salary Question

| emp_name | department | salary |
|----------|------------|--------|
| AMIT     | IT         | 50000  |
| ROHIT    | HR         | 60000  |
| NEHA     | IT         | 70000  |
| PRIYA    | HR         | 80000  |
| RAVI     | SALES      | 40000  |

### Department-wise Total Salary

```
SELECT department, SUM(salary) AS total_salary
FROM employees GROUP BY department;
```

| department | total_salary |
|------------|--------------|
| HR         | 140000       |
| IT         | 120000       |
| SALES      | 40000        |

### Highest Salary ka Department kaun sa hai?

```
SELECT department, SUM(salary) AS total_salary FROM employees
GROUP BY department
ORDER BY total_salary DESC
LIMIT 1;
```

| department | total_salary |
|------------|--------------|
| HR         | 140000       |

**Department Having Total Salary > 100000**

```
SELECT department, SUM(salary) AS total_salary
FROM employees GROUP BY department
HAVING SUM(salary) > 100000;
```

| department | total_salary |
|------------|--------------|
| HR         | 140000       |
| IT         | 120000       |

**Department-wise average salary?**

```
SELECT department, AVG(salary) AS avg_salary
FROM employees GROUP BY department;
```

| department | avg_salary |
|------------|------------|
| HR         | 70000      |
| IT         | 60000      |
| SALES      | 45000      |



## Count email number

```
+-----+-----+-----+
| id | emp_name | email          |
+-----+-----+-----+
| 1 | Mohit    | mohit@gmail.com |
| 2 | Rahul    | rahul@gmail.com |
| 3 | Amit     | mohit@gmail.com |
| 4 | Sumit    | NULL           |
| 5 | Raj      | rahul@gmail.com |
+-----+-----+-----+
```

- Total non-NULL emails

```
SELECT COUNT(email) AS total_email FROM Person;
```

```
+-----+
| total_email |
+-----+
|           4 |
+-----+
```

- Count each email

```
SELECT email, COUNT(email) AS total_email FROM Person GROUP BY email;
```

```
+-----+-----+
| email          | total_email |
+-----+-----+
| NULL          |           0 |
| mohit@gmail.com |           2 |
| rahul@gmail.com |           2 |
+-----+-----+
```

- Only duplicate emails

```
SELECT email, COUNT(email) AS total_email FROM Person
GROUP BY email HAVING COUNT(email) > 1;
```

```
+-----+-----+
| email          | total_email |
+-----+-----+
| mohit@gmail.com |           2 |
+-----+-----+
```

```
| rahul@gmail.com | 2 |  
+-----+-----+
```

## 🎯 How to Find Duplicate values in a Table?

| id | emp_name | email           |
|----|----------|-----------------|
| 1  | Mohit    | mohit@gmail.com |
| 2  | Rahul    | rahul@gmail.com |
| 3  | Amit     | mohit@gmail.com |
| 4  | Sumit    | sumit@gmail.com |
| 5  | Raj      | rahul@gmail.com |

```
-- COUNT(*) Null value ko bhi count karta hai
SELECT email, COUNT(*) AS count FROM employees GROUP BY email
HAVING COUNT(*) > 1;

-- COUNT(Email) NULL valu ko count nahi karta hai
SELECT Email, COUNT(Email) AS total_email FROM employees GROUP BY Email
HAVING COUNT(Email) > 1;
```

| Email           | total_email |
|-----------------|-------------|
| mohit@gmail.com | 2           |
| rahul@gmail.com | 2           |

```
-- NULL values mtb
```

| email           |
|-----------------|
| mohit@gmail.com |
| rahul@gmail.com |
| NULL            |
| sumit@gmail.com |
| rahul@gmail.com |

```
-- If you want the complete records of duplicates
SELECT *
FROM employees
WHERE email IN (
    SELECT email
    FROM employees
    GROUP BY email
    HAVING COUNT(*) > 1
);
```

## Duplicate value remove

- Using Join

```
DELETE t1
FROM my_table t1
INNER JOIN my_table t2
    ON t1.col1 = t2.col1
    AND t1.col2 = t2.col2 -- agar aap duplicate ko multiple columns ke combination
ke basis par identify karna chahte ho, tab use karoge.
    AND t1.id > t2.id;
```

- Using Subquery

```
DELETE FROM Person
WHERE id IN (
    SELECT id
    FROM (
        SELECT t1.id
        FROM Person t1
        JOIN Person t2
            ON t1.email = t2.email
            AND t1.id > t2.id
        ) AS temp
);
```

## Replace a Column Values from 'male' to 'female' and 'female' to 'male'

```
UPDATE empdata
SET GENDER = CASE
    WHEN GENDER='male' THEN 'female'
    WHEN GENDER='female' THEN 'male'
END;

(OR)

UPDATE EMPDATA
SET gender = CASE
    gender WHEN 'male' THEN 'female'
           WHEN 'female' THEN 'male'
    ELSE gender
END;
```